

Mahendra Vaka

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PROFESSIONAL SUMMARY

Results-driven Data Analyst with 4+ years of experience in financial services and healthcare analytics, specializing in data pipeline development, business intelligence, and predictive modeling. Expertise in SQL, Python, Tableau, and Azure cloud technologies, with proven track record of processing \$50B+ daily transaction volumes and improving operational efficiency by 70% through automation. Seeking to leverage advanced analytical skills and machine learning expertise to drive data-informed decision making and operational excellence for enterprise-level organizations.

EDUCATION

Trine University

Master of Science in Business Analytics

Detroit, MI

May 2024

VVIT

Bachelor of Engineering in Electrical and Electronics Engineering

Guntur, India

August 2021

PROFESSIONAL EXPERIENCE

Northern Trust

Data Analyst

Dallas, TX

January 2023 – May 2025

- Architected comprehensive data pipeline infrastructure processing \$50B+ daily transaction volumes across 15+ automated systems, achieving 99.9% data accuracy and reliability through advanced ETL automation and quality validation frameworks.
- Engineered integrated Tableau dashboard ecosystem consolidating multiple data sources for portfolio management teams, delivering real-time risk assessment insights through advanced data visualization and business intelligence architecture.
- Implemented enterprise data governance framework across 75+ heterogeneous data sources, reducing data inconsistencies by 25% and establishing standardized metadata practices through Collibra and Microsoft Purview integration.
- Developed automated validation and reporting workflows streamlining monthly reporting cycles, reducing processing time by 40% and eliminating 120+ hours of manual work through Python automation and CI/CD pipeline implementation.
- Spearheaded predictive analytics model development using Python and R for comprehensive risk assessment, improving prediction accuracy by 30% across multiple asset classes through machine learning algorithms and statistical modeling frameworks.

Mphasis

Junior Data Analyst

Bangalore, India

January 2020 – July 2022

- Orchestrated large-scale enterprise data analysis initiative processing 1TB+ data from multiple business systems, creating 25+ optimized data marts and reducing analysis time by 65% through advanced data architecture and optimization techniques.
- Designed comprehensive business intelligence dashboard ecosystem tracking operational efficiency and quality metrics across departments, processing 100+ daily data files with 99.8% accuracy through KPI development and visualization frameworks.
- Developed machine learning models for business forecasting implementing advanced statistical algorithms and OLAP functions, improving operational planning efficiency by 25% and enabling predictive outcome analysis through Python-based predictive analytics and statistical modeling.
- Engineered automated Python integration pipelines connecting multiple enterprise systems and standardizing 300+ business data sources, eliminating data silos and improving integration efficiency through ETL automation and system integration frameworks.

PROJECTS

Sales Forecast Analysis – CSU Football Magazine

- Utilized regression and simulation models to forecast Year 10 magazine sales, improving prediction accuracy by deselecting irrelevant variables and quantifying key sales drivers using historical data through advanced statistical modeling techniques.
- Developed strategic inventory and newsvendor models to minimize lost sales and excess stock costs, optimizing marketing, production, and distribution resources through comprehensive supply chain analysis and cost modeling.

Data-Driven Online Bookstore Strategy – The Bookworm Shack

- Leveraged digital analytics to analyze customer browsing and purchase patterns, designing personalized recommendations and targeted promotions that increased satisfaction and boosted conversion rates through data-driven customer segmentation strategies.
- Integrated customer feedback into data-driven decisions, improving engagement and sales and strategic marketing effectiveness through continuous insight-driven enhancements and performance optimization methodologies.

Retirement Planning Financial Model – Bob Case

- Built comprehensive retirement model forecasting optimal annual savings, investment mix, and risk mitigation plans to secure comfortable post-retirement lifestyle for hypothetical client through advanced financial planning and modeling techniques.
- Analyzed income growth, asset returns, and retirement costs, delivering actionable recommendations and tailored stock investment advice to balance living standards with financial risks through comprehensive risk assessment methodologies.

Financial Status Analysis – Petróleo Brasileiro (Petrobras)

- Analyzed Petrobras's 2020–2021 balance sheets to assess changes in key metrics, identifying 18.9% rise in current liabilities and 775.7% increase in accounts payable, highlighting significant financial shifts through comprehensive financial statement analysis.
- Interpreted financial trends, highlighting liquidity risks from 5.8% cash decline and advising strategic improvements in liability management and asset utilization to strengthen financial stability and performance through strategic financial consulting.

TECHNICAL SKILLS

Programming & Analytics: Python (pandas, NumPy, scikit-learn, scipy), R, SQL (Expert), SAS, Statistical Analysis, Machine Learning, Predictive Analytics

Data Visualization & BI: Tableau (Expert), Power BI, Advanced Excel, Dashboard Development, KPI Development, Business Reporting

Data Engineering: Azure Data Factory, SQL Server, Teradata, Data Pipeline Development, ETL Processes, Data Quality Management, Apache Spark

Cloud & DevOps: Azure PaaS, Azure Functions, Microservices, CI/CD Pipelines, GitHub, Azure DevOps, SonarQube

Analytics & Methodologies: Time Series Analysis, A/B Testing, Hypothesis Testing, Agile/SAFe, Lean Six Sigma, DMAIC, Risk Assessment

Data Management: Collibra, Informatica, Microsoft Purview, Data Cataloging, Lineage Mapping, Metadata Management